Pikachu Metrics 数据分析 metrics_Mjlab-Tracking-Flat-Pikachu-Down__model_4999_20260820_103311

FILE
metrics_Mjlab-Tracking-Flat-Pikachu-Down__model_4999_20260820_103311.csv

NETWORK
Mjlab-Tracking-Flat-Pikachu-Down__model_4999

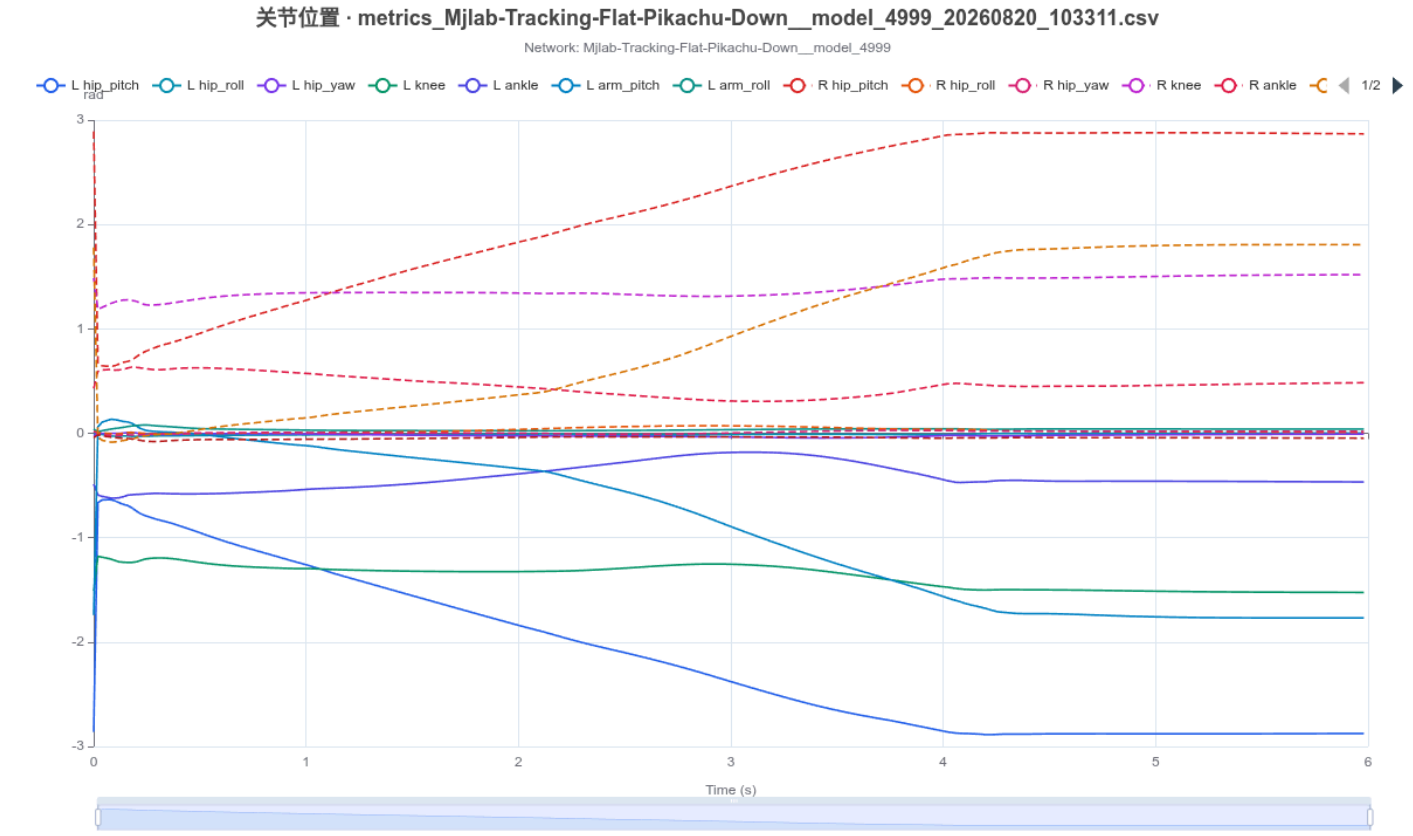
SAMPLES
277

DURATION
5.980 s

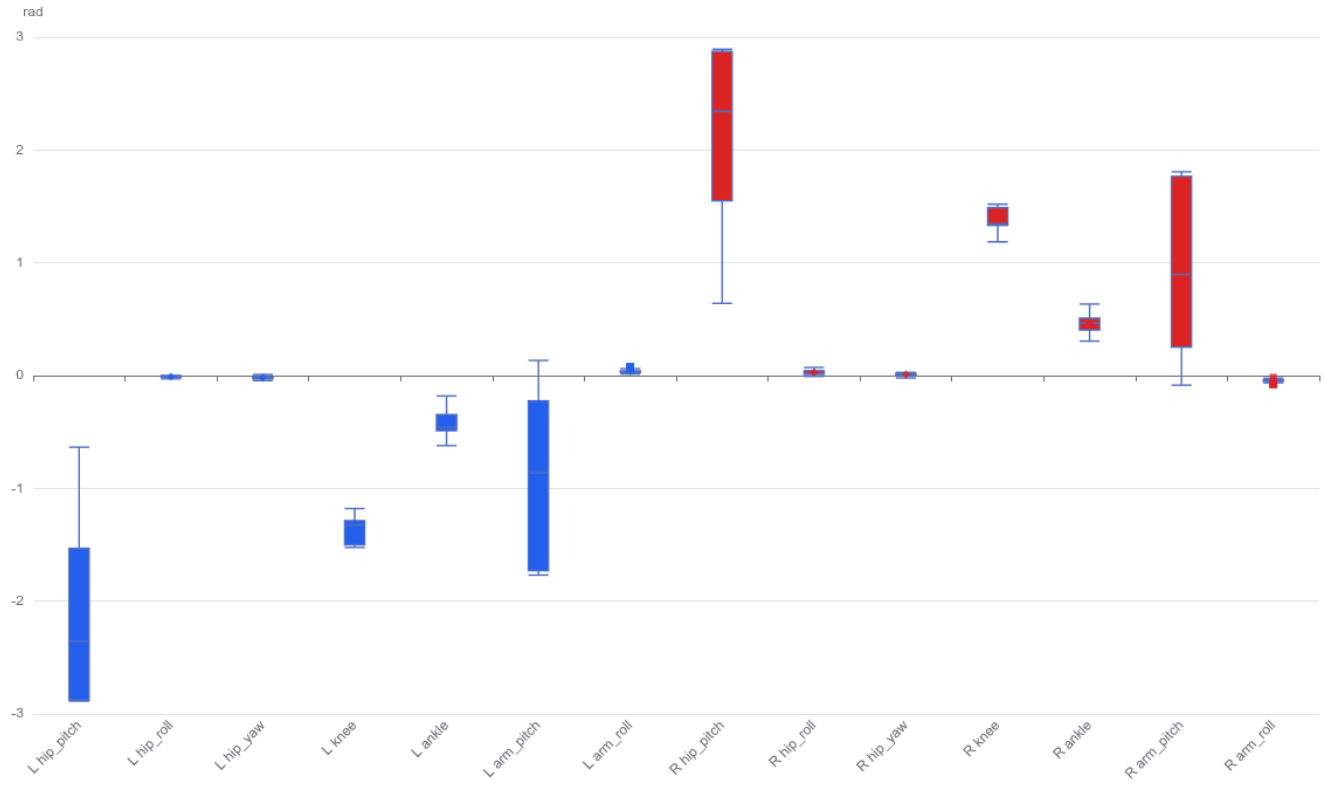
ENVIRONMENTS
1

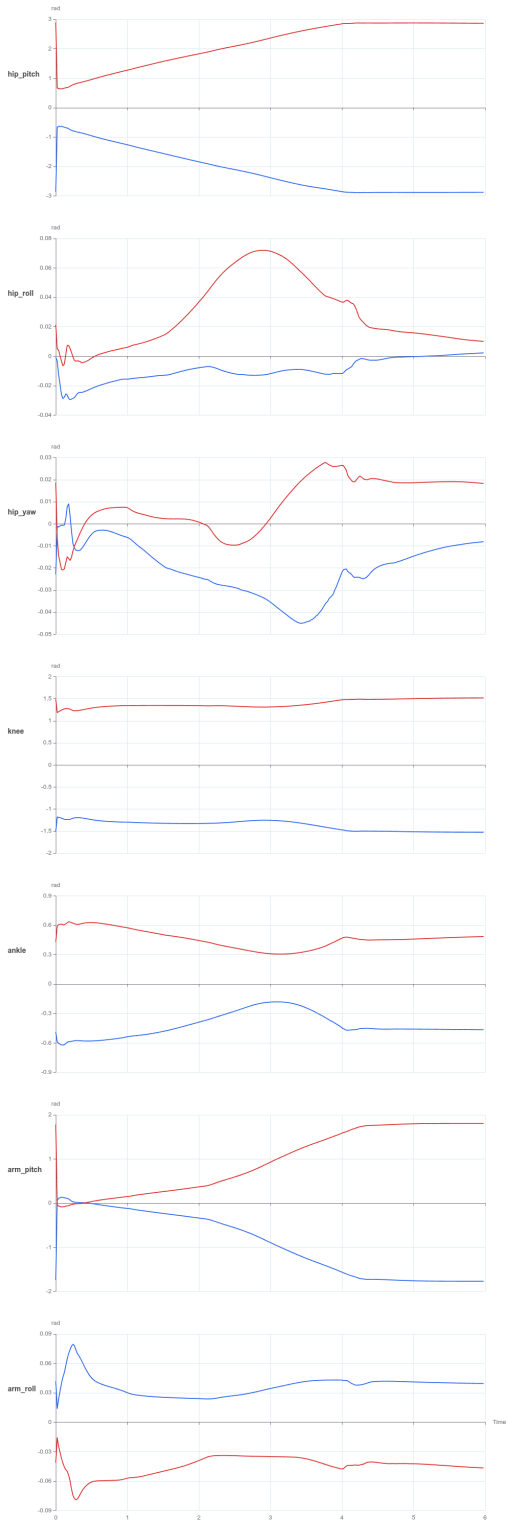
时间区间: 0.000 — 5.980 s · 仅含已勾选字段

关节位置

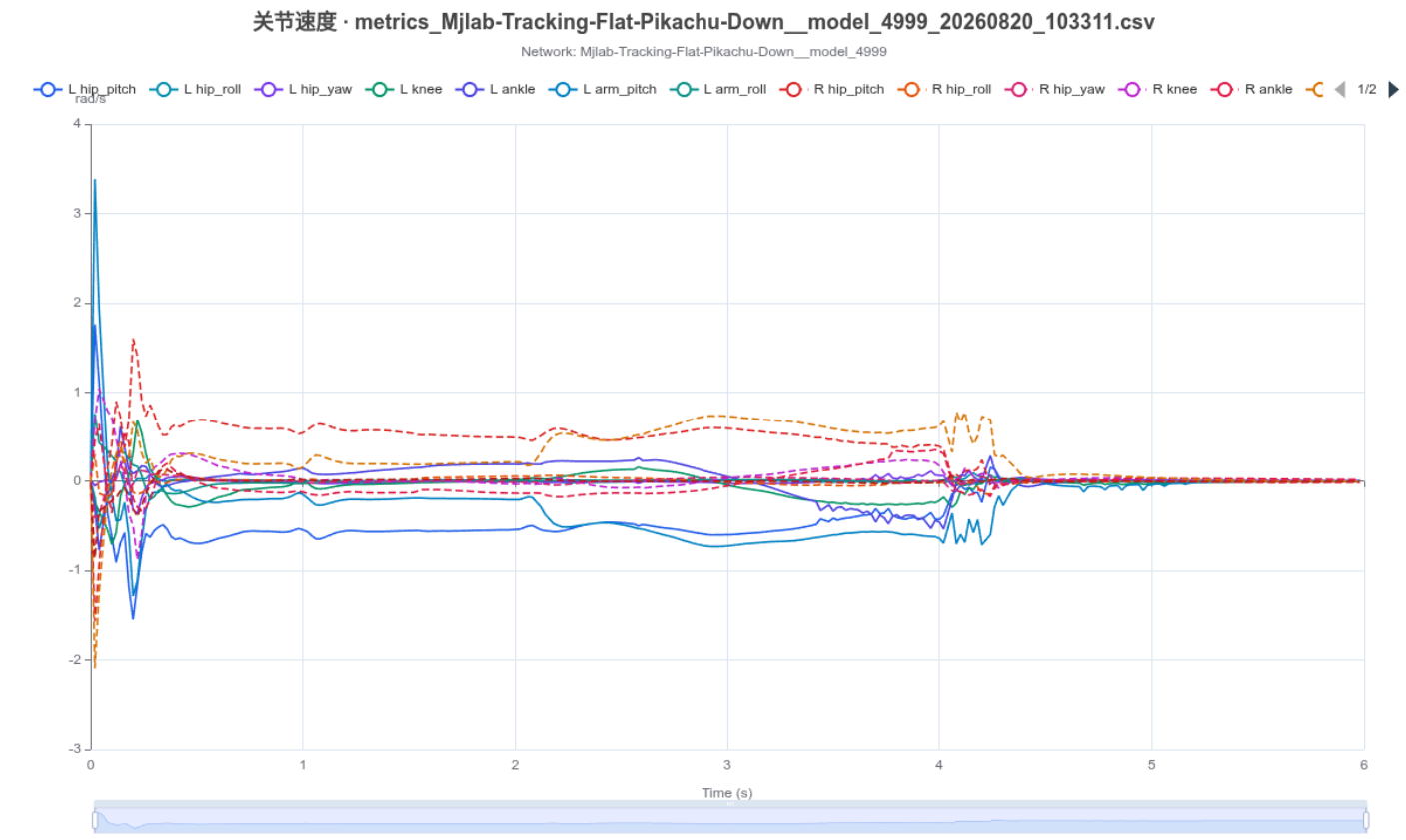


Metric	Vel限	Torq限	N	Min	Max	Mean	RMS	Peak |x|	P95 |x|
joint_position_rad/left_hip_pitch_joint	30	10	277	-2.88656	-0.635259	-2.15691	2.27820	2.88656	2.87925
joint_position_rad/left_hip_roll_joint	10	15	277	-0.0295424	0.00227633	-0.00949265	0.0121831	0.0295424	0.0246191
joint_position_rad/left_hip_yaw_joint	10	15	277	-0.0450674	0.00909267	-0.0198234	0.0231838	0.0450674	0.0424743
joint_position_rad/left_knee_joint	30	10	277	-1.52409	-1.17889	-1.37007	1.37443	1.52409	1.52261
joint_position_rad/left_ankle_joint	10	15	277	-0.621231	-0.181415	-0.422860	0.439338	0.621231	0.580322
joint_position_rad/left_arm_pitch_joint	10	15	277	-1.76893	0.133848	-0.915305	1.15514	1.76893	1.76862
joint_position_rad/left_arm_roll_joint	10	15	277	0.0137871	0.0794785	0.0369538	0.0383533	0.0794785	0.0540010
joint_position_rad/right_hip_pitch_joint	30	10	277	0.639974	2.89235	2.15519	2.27508	2.89235	2.87897
joint_position_rad/right_hip_roll_joint	10	15	277	-0.00655684	0.0719218	0.0272944	0.0356435	0.0719218	0.0704031
joint_position_rad/right_hip_yaw_joint	10	15	277	-0.0209241	0.0277465	0.00911929	0.0148643	0.0277465	0.0260216
joint_position_rad/right_knee_joint	30	10	277	1.18600	1.51954	1.39079	1.39351	1.51954	1.51648
joint_position_rad/right_ankle_joint	10	15	277	0.305651	0.634090	0.462993	0.471867	0.634090	0.621056
joint_position_rad/right_arm_pitch_joint	10	15	277	-0.0852998	1.80695	0.947469	1.18259	1.80695	1.80680
joint_position_rad/right_arm_roll_joint	10	15	277	-0.0790789	-0.0156294	-0.0451134	0.0461330	0.0790789	0.0604941

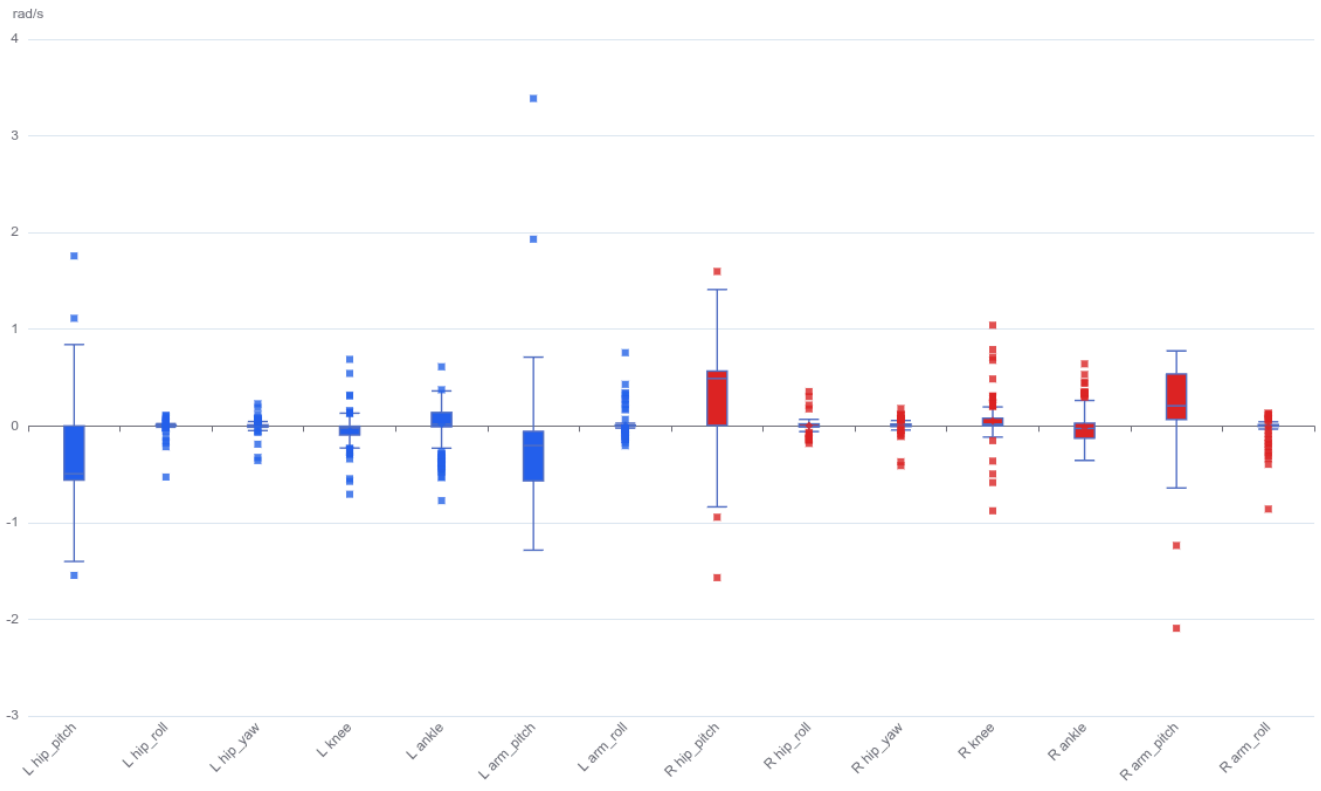

rad
3
2
1
0
-1
-2
-3
L hip_pitch
L hip_roll
L hip_yaw
L knee
L ankle
L arm_pitch
L arm_roll
R hip_pitch
R hip_roll
R hip_yaw
R knee
R ankle
R arm_pitch
R arm_roll

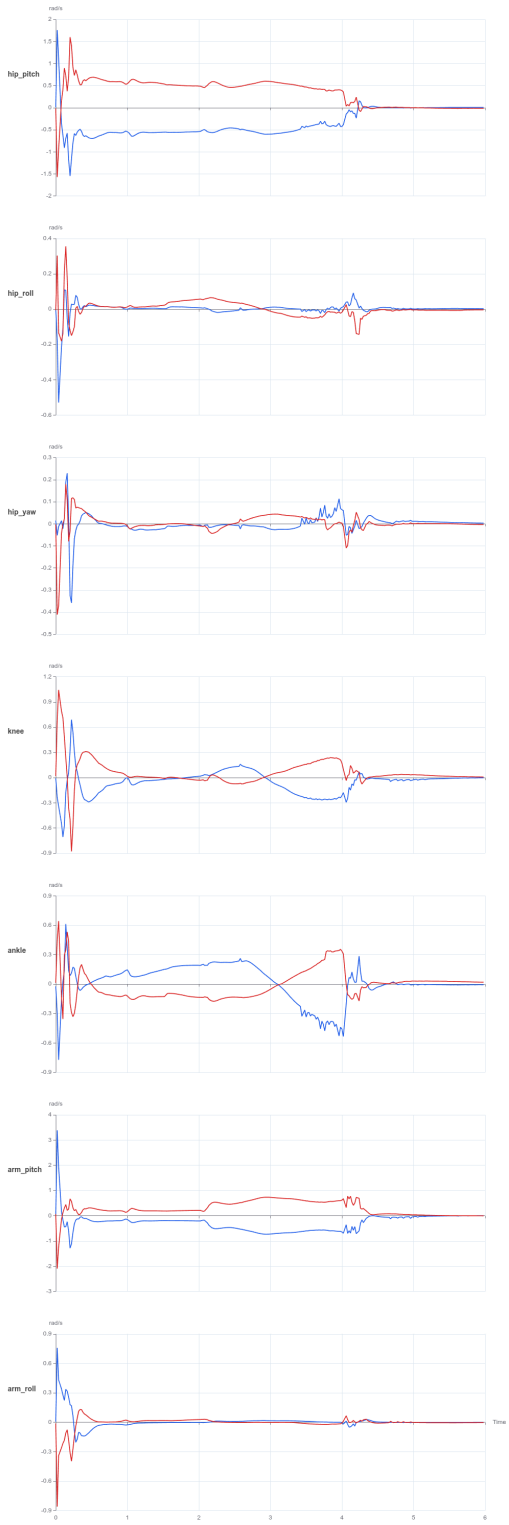

hip_pitch
hip_roll
hip_yaw
knee
ankle
arm_pitch
arm_roll
rad
Time

关节速度

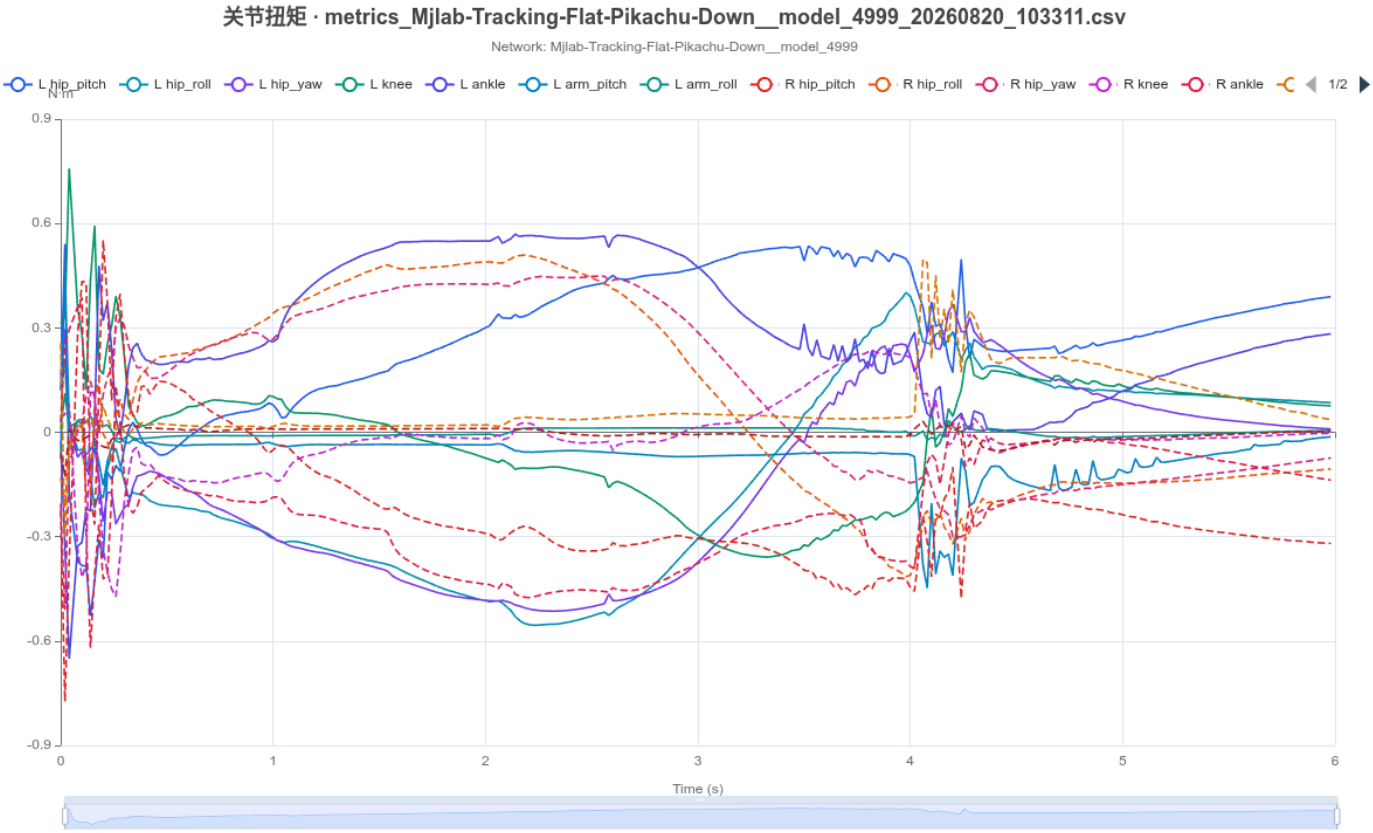


Metric	Vel限	Torq限	N	Min	Max	Mean	RMS	Peak |x|	P95 |x|
joint_velocity_rad_s/left_hip_pitch_joint	30	10	277	-1.54637	1.75805	-0.356689	0.478441	1.75805	0.659036
joint_velocity_rad_s/left_hip_roll_joint	10	15	277	-0.528889	0.108796	0.00144210	0.0413961	0.528889	0.0407462
joint_velocity_rad_s/left_hip_yaw_joint	10	15	277	-0.358360	0.229024	-0.00175909	0.0426256	0.358360	0.0522575
joint_velocity_rad_s/left_knee_joint	30	10	277	-0.706632	0.688402	-0.0520867	0.146499	0.706632	0.264354
joint_velocity_rad_s/left_ankle_joint	10	15	277	-0.772788	0.611684	0.0211041	0.180661	0.772788	0.383828
joint_velocity_rad_s/left_arm_pitch_joint	10	15	277	-1.28541	3.38705	-0.274285	0.455148	3.38705	0.708057
joint_velocity_rad_s/left_arm_roll_joint	10	15	277	-0.204876	0.757242	0.00544737	0.0758451	0.757242	0.137181
joint_velocity_rad_s/right_hip_pitch_joint	30	10	277	-1.56966	1.59748	0.356698	0.474769	1.59748	0.679123
joint_velocity_rad_s/right_hip_roll_joint	10	15	277	-0.181223	0.355566	0.00291062	0.0500856	0.355566	0.0669405
joint_velocity_rad_s/right_hip_yaw_joint	10	15	277	-0.411480	0.179774	0.00346917	0.0444108	0.411480	0.0688796
joint_velocity_rad_s/right_knee_joint	30	10	277	-0.878051	1.04160	0.0514180	0.159496	1.04160	0.276724
joint_velocity_rad_s/right_ankle_joint	10	15	277	-0.356841	0.641735	-0.0171473	0.145766	0.641735	0.332083
joint_velocity_rad_s/right_arm_pitch_joint	10	15	277	-2.09342	0.775357	0.280454	0.406681	2.09342	0.710554
joint_velocity_rad_s/right_arm_roll_joint	10	15	277	-0.860087	0.129504	-0.00734367	0.0741634	0.860087	0.101952

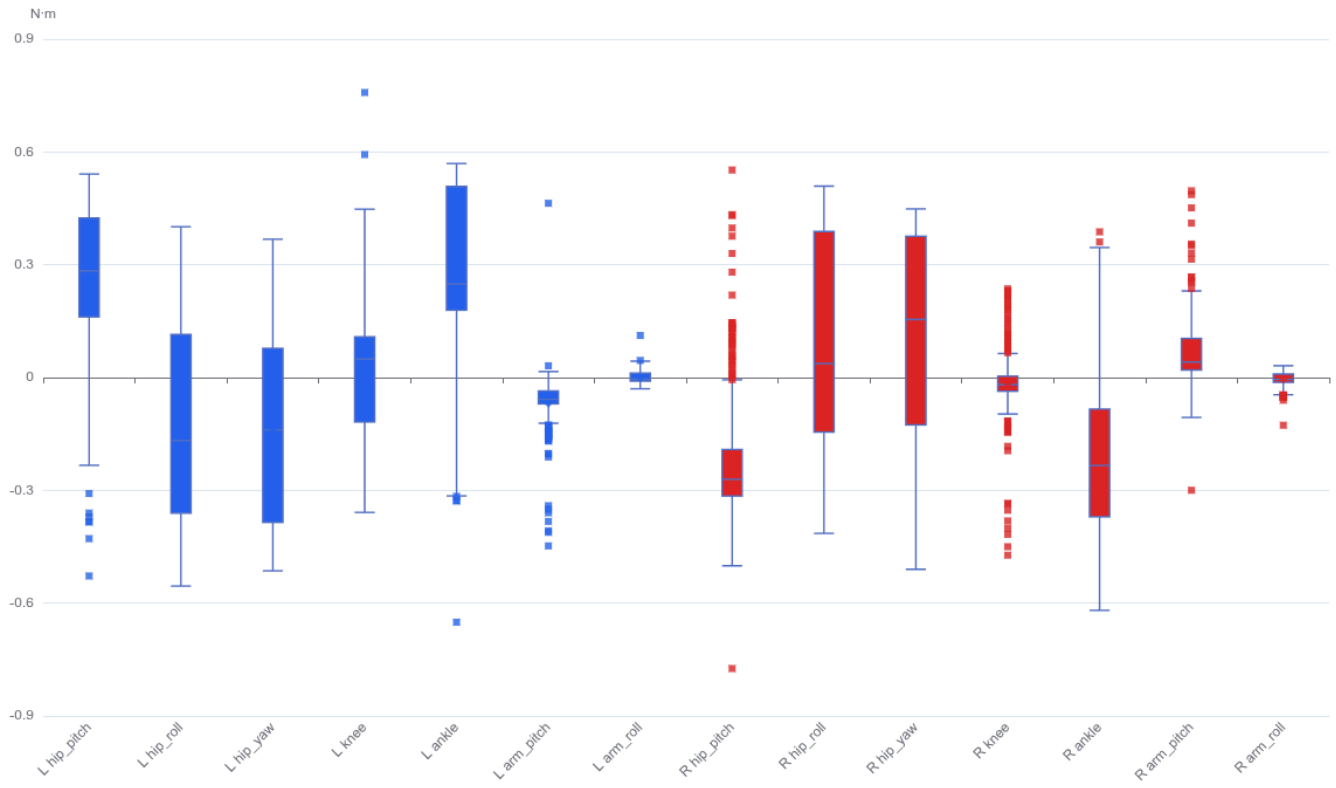

rad/s
4
3
2
1
0
-1
-2
-3
L hip_pitch
L hip_roll
L hip_yaw
L knee
L ankle
L arm_pitch
L arm_roll
R hip_pitch
R hip_roll
R hip_yaw
R knee
R ankle
R arm_pitch
R arm_roll

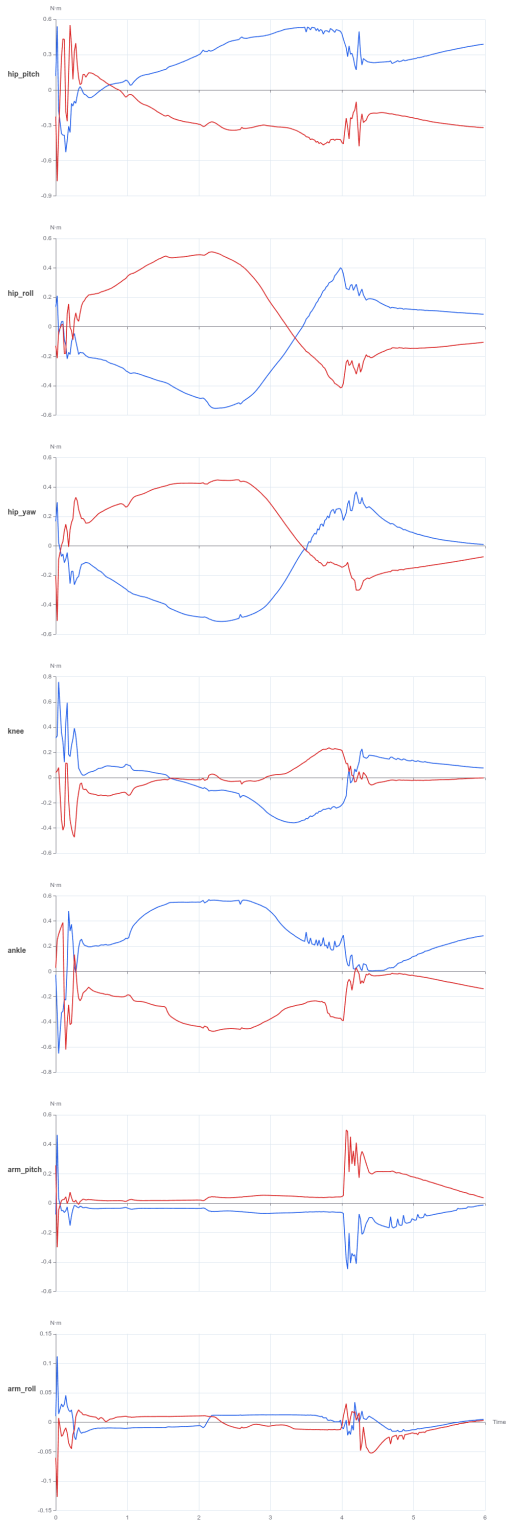

rad/s
hip_pitch
rad/s
hip_roll
rad/s
hip_yaw
rad/s
knee
rad/s
ankle
rad/s
arm_pitch
rad/s
arm_roll
Time

关节扭矩

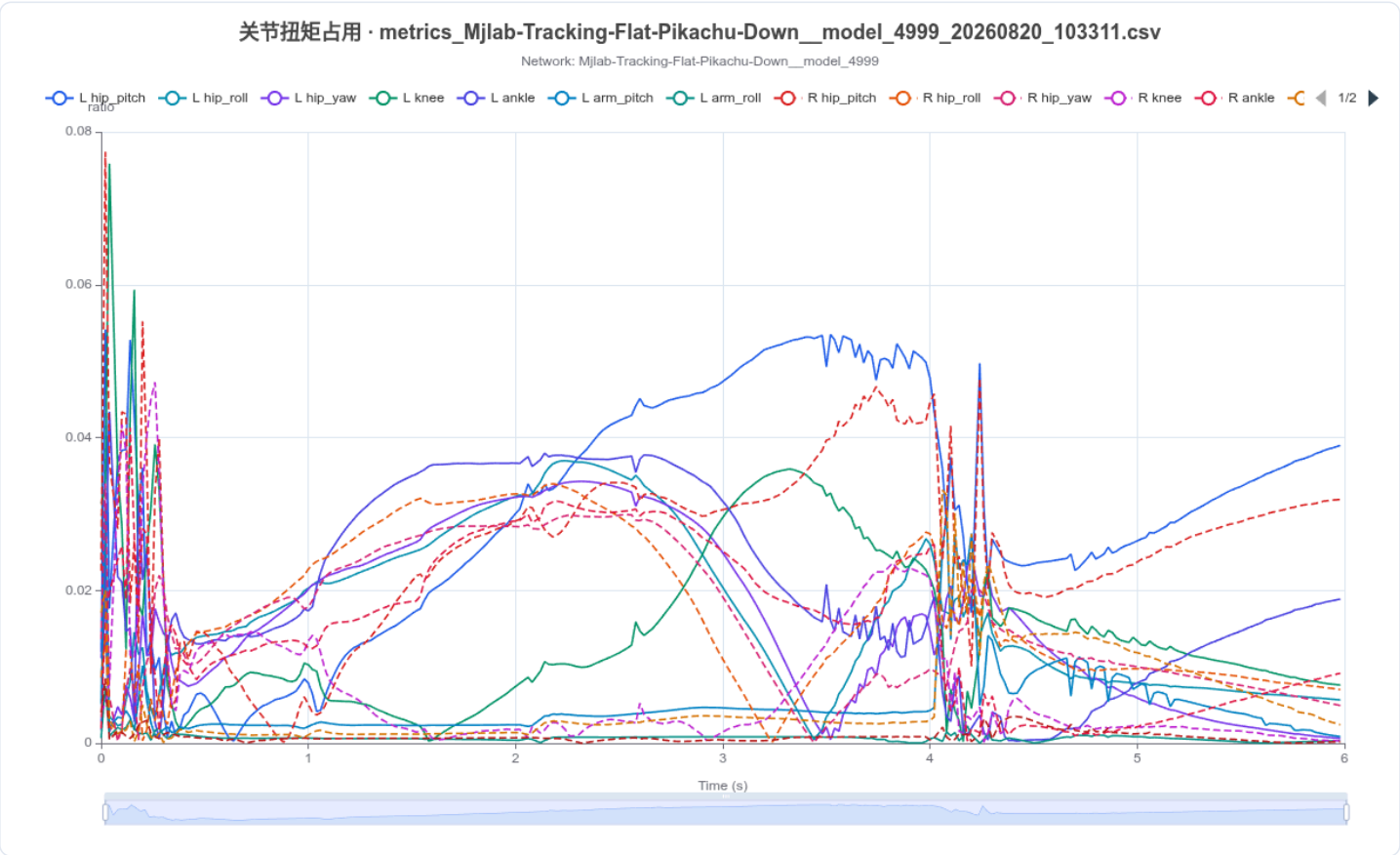


Metric	Vel限	Torq限	N	Min	Max	Mean	RMS	Peak |x|	P95 |x|
joint_torque_nm/left_hip_pitch_joint	30	10	277	-0.527728	0.541344	0.267502	0.332964	0.541344	0.522729
joint_torque_nm/left_hip_roll_joint	10	15	277	-0.554466	0.401189	-0.121049	0.297689	0.554466	0.535322
joint_torque_nm/left_hip_yaw_joint	10	15	277	-0.513990	0.367906	-0.138374	0.294189	0.513990	0.501230
joint_torque_nm/left_knee_joint	30	10	277	-0.358722	0.758239	-0.00667525	0.181002	0.758239	0.348754
joint_torque_nm/left_ankle_joint	10	15	277	-0.650698	0.569251	0.286697	0.352898	0.650698	0.561303
joint_torque_nm/left_arm_pitch_joint	10	15	277	-0.447790	0.463613	-0.0686804	0.0985626	0.463613	0.161719
joint_torque_nm/left_arm_roll_joint	10	15	277	-0.0296502	0.111980	-0.0000524623	0.0133213	0.111980	0.0188119
joint_torque_nm/right_hip_pitch_joint	30	10	277	-0.773923	0.552088	-0.217578	0.280795	0.773923	0.436498
joint_torque_nm/right_hip_roll_joint	10	15	277	-0.414475	0.509197	0.0916693	0.298553	0.509197	0.486760
joint_torque_nm/right_hip_yaw_joint	10	15	277	-0.509971	0.448660	0.116876	0.274943	0.509971	0.444367
joint_torque_nm/right_knee_joint	30	10	277	-0.472372	0.236010	-0.0165190	0.108091	0.472372	0.224619
joint_torque_nm/right_ankle_joint	10	15	277	-0.619158	0.387676	-0.222154	0.277231	0.619158	0.456173
joint_torque_nm/right_arm_pitch_joint	10	15	277	-0.299741	0.496786	0.0768110	0.118622	0.496786	0.217837
joint_torque_nm/right_arm_roll_joint	10	15	277	-0.126839	0.0315603	-0.00544383	0.0182216	0.126839	0.0404288

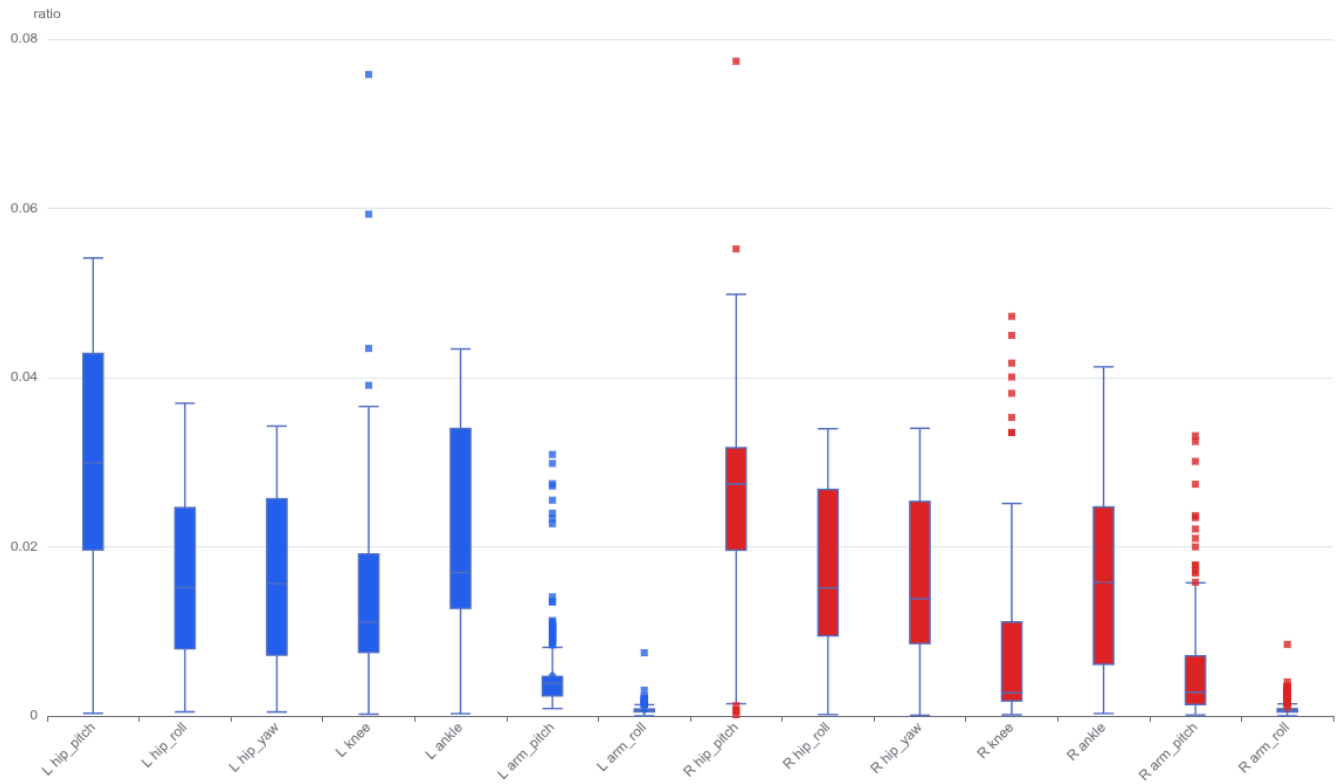

N·m
0.9
0.6
0.3
0
-0.3
-0.6
-0.9
L hip_pitch
L hip_roll
L hip_yaw
L knee
L ankle
L arm_pitch
L arm_roll
R hip_pitch
R hip_roll
R hip_yaw
R knee
R ankle
R arm_pitch
R arm_roll

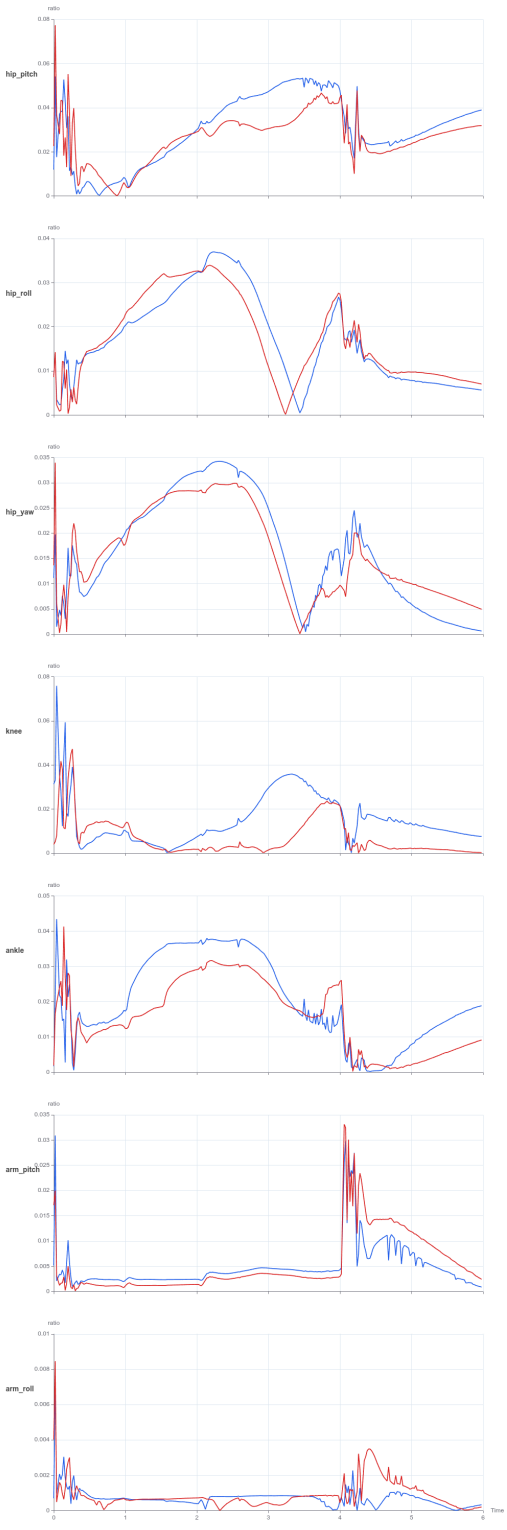

hip_pitch
hip_roll
hip_yaw
knee
ankle
arm_pitch
arm_roll
Time

关节扭矩占用



Metric	Vel限	Torq限	N	Min	Max	Mean	RMS	Peak |x|	P95 |x|
joint_torque_ratio/left_hip_pitch_joint	30	10	277	0.000310152	0.0541344	0.0295744	0.0332964	0.0541344	0.0522729
joint_torque_ratio/left_hip_roll_joint	10	15	277	0.000484302	0.0369644	0.0171748	0.0198460	0.0369644	0.0356881
joint_torque_ratio/left_hip_yaw_joint	10	15	277	0.000465698	0.0342660	0.0164364	0.0196126	0.0342660	0.0334153
joint_torque_ratio/left_knee_joint	30	10	277	0.000206099	0.0758239	0.0145629	0.0181002	0.0758239	0.0348754
joint_torque_ratio/left_ankle_joint	10	15	277	0.000266936	0.0433799	0.0201256	0.0235265	0.0433799	0.0374202
joint_torque_ratio/left_arm_pitch_joint	10	15	277	0.000869188	0.0309075	0.00481691	0.00657084	0.0309075	0.0107813
joint_torque_ratio/left_arm_roll_joint	10	15	277	0.00000536377	0.00746532	0.000694796	0.000888087	0.00746532	0.00125413
joint_torque_ratio/right_hip_pitch_joint	30	10	277	0.000133787	0.0773923	0.0257162	0.0280795	0.0773923	0.0436498
joint_torque_ratio/right_hip_roll_joint	10	15	277	0.000170542	0.0339465	0.0174163	0.0199035	0.0339465	0.0324506
joint_torque_ratio/right_hip_yaw_joint	10	15	277	0.0000778354	0.0339981	0.0160709	0.0183295	0.0339981	0.0296245
joint_torque_ratio/right_knee_joint	30	10	277	0.000137730	0.0472372	0.00686616	0.0108091	0.0472372	0.0224619
joint_torque_ratio/right_ankle_joint	10	15	277	0.000278636	0.0412772	0.0155352	0.0184821	0.0412772	0.0304116
joint_torque_ratio/right_arm_pitch_joint	10	15	277	0.000106290	0.0331191	0.00529291	0.00790811	0.0331191	0.0145225
joint_torque_ratio/right_arm_roll_joint	10	15	277	0.00000461696	0.00845591	0.000879796	0.00121478	0.00845591	0.00269525

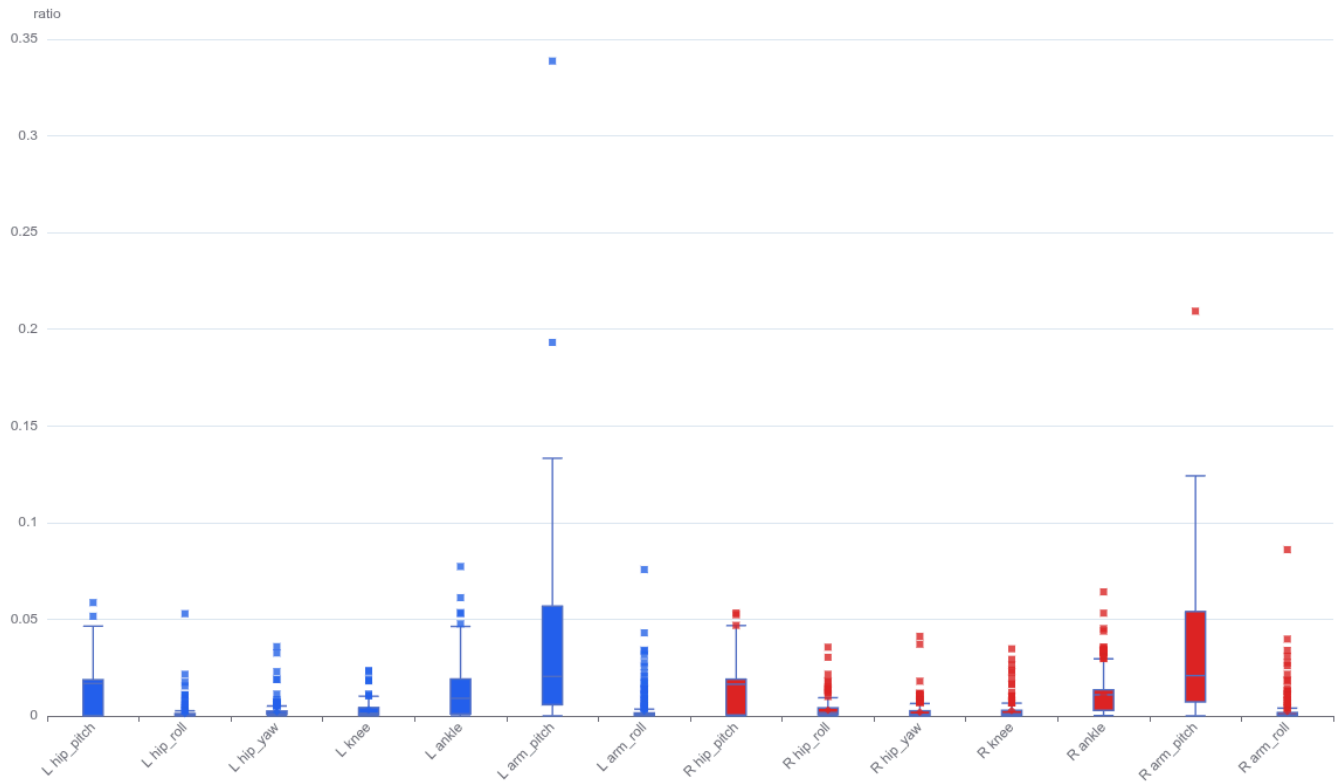

ratio
0.08
0.06
0.04
0.02
0
L hip_pitch
L hip_roll
L hip_yaw
L knee
L ankle
L arm_pitch
L arm_roll
R hip_pitch
R hip_roll
R hip_yaw
R knee
R ankle
R arm_pitch
R arm_roll

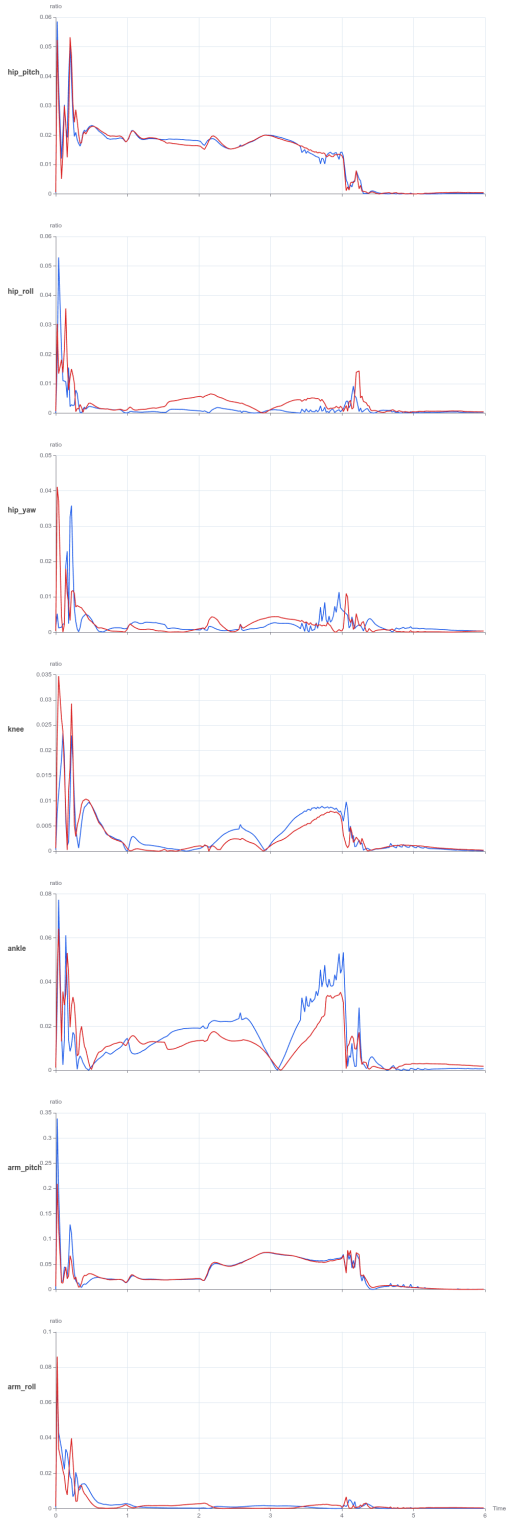

hip_pitch
hip_roll
hip_yaw
knee
ankle
arm_pitch
arm_roll
Time

关节速度占用

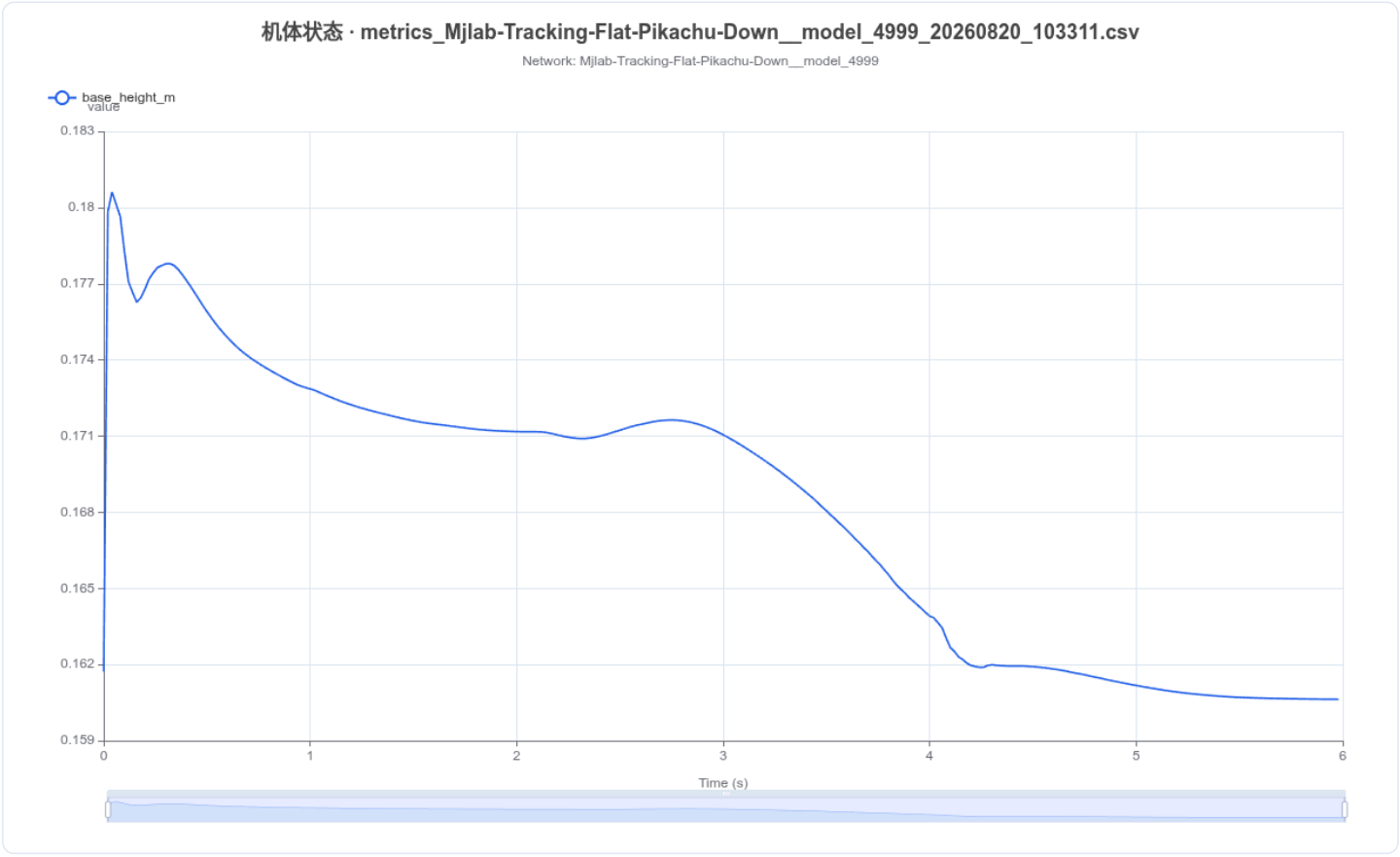


Metric	Vel限	Torq限	N	Min	Max	Mean	RMS	Peak |x|	P95 |x|
joint_velocity_ratio/left_hip_pitch_joint	30	10	277	0.00000226851	0.0586017	0.0127483	0.0159480	0.0586017	0.0219679
joint_velocity_ratio/left_hip_roll_joint	10	15	277	0.00000336304	0.0528889	0.00139330	0.00413961	0.0528889	0.00407462
joint_velocity_ratio/left_hip_yaw_joint	10	15	277	0.0000209352	0.0358360	0.00215765	0.00426256	0.0358360	0.00522575
joint_velocity_ratio/left_knee_joint	30	10	277	0.0000232114	0.0235544	0.00310480	0.00488331	0.0235544	0.00881179
joint_velocity_ratio/left_ankle_joint	10	15	277	0.0000204815	0.0772788	0.0126910	0.0180661	0.0772788	0.0383828
joint_velocity_ratio/left_arm_pitch_joint	10	15	277	0.0000123997	0.338705	0.0313894	0.0455148	0.338705	0.0708057
joint_velocity_ratio/left_arm_roll_joint	10	15	277	0.0000142008	0.0757242	0.00258163	0.00758451	0.0757242	0.0137181
joint_velocity_ratio/right_hip_pitch_joint	30	10	277	0.00000710528	0.0532494	0.0126854	0.0158256	0.0532494	0.0226374
joint_velocity_ratio/right_hip_roll_joint	10	15	277	0.0000231848	0.0355566	0.00300638	0.00500856	0.0355566	0.00669405
joint_velocity_ratio/right_hip_yaw_joint	10	15	277	0.00000169805	0.0411480	0.00200648	0.00444108	0.0411480	0.00688796
joint_velocity_ratio/right_knee_joint	30	10	277	3.78100e-7	0.0347201	0.00280499	0.00531652	0.0347201	0.00922414
joint_velocity_ratio/right_ankle_joint	10	15	277	0.000167914	0.0641735	0.0109308	0.0145766	0.0641735	0.0332083
joint_velocity_ratio/right_arm_pitch_joint	10	15	277	0.0000208528	0.209342	0.0305534	0.0406681	0.209342	0.0710554
joint_velocity_ratio/right_arm_roll_joint	10	15	277	0.0000186795	0.0860087	0.00237879	0.00741635	0.0860087	0.0101952

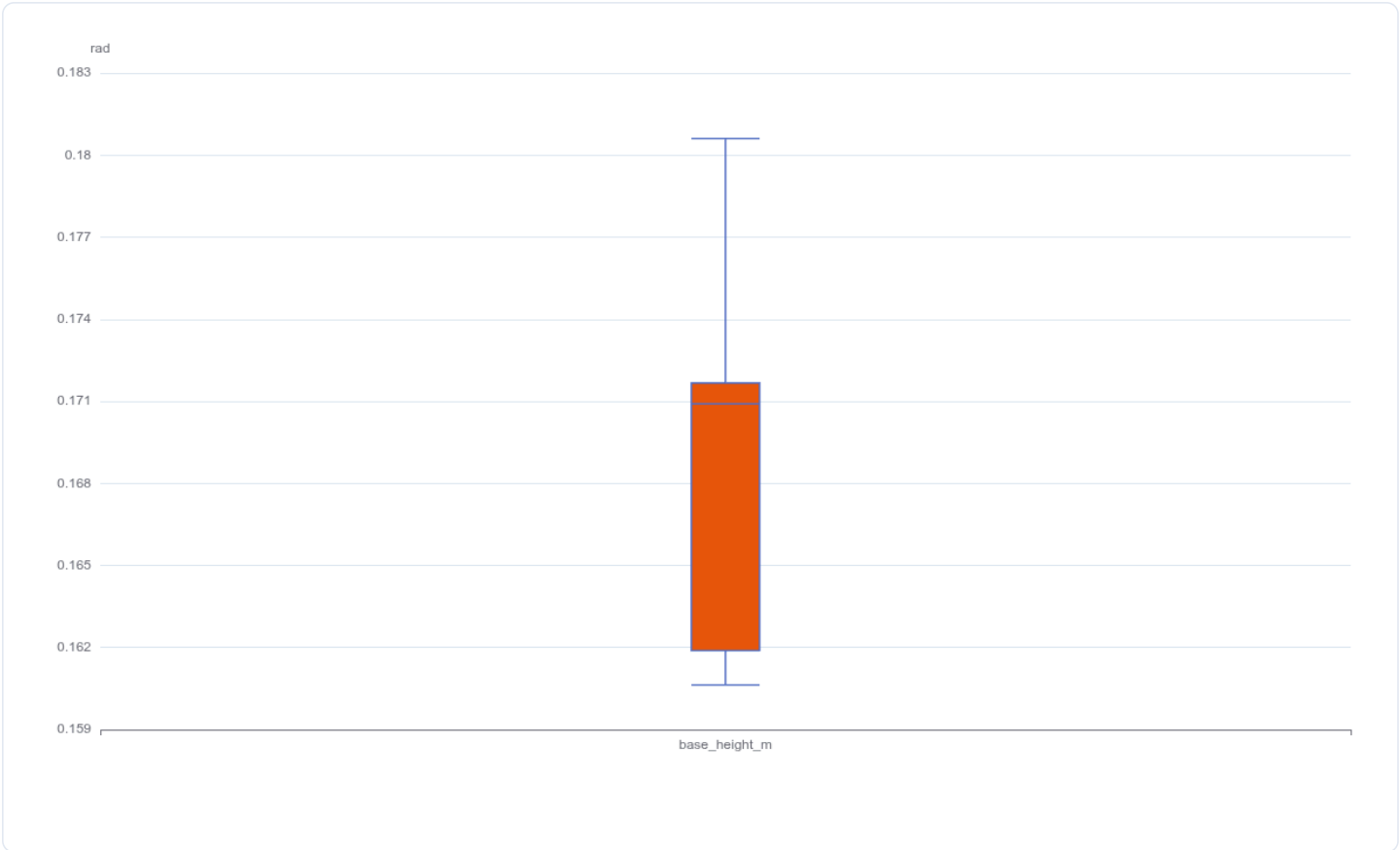

ratio
0.35
0.3
0.25
0.2
0.15
0.1
0.05
0
L hip_pitch
L hip_roll
L hip_yaw
L knee
L ankle
L arm_pitch
L arm_roll
R hip_pitch
R hip_roll
R hip_yaw
R knee
R ankle
R arm_pitch
R arm_roll


hip_pitch
hip_roll
hip_yaw
knee
ankle
arm_pitch
arm_roll
Time

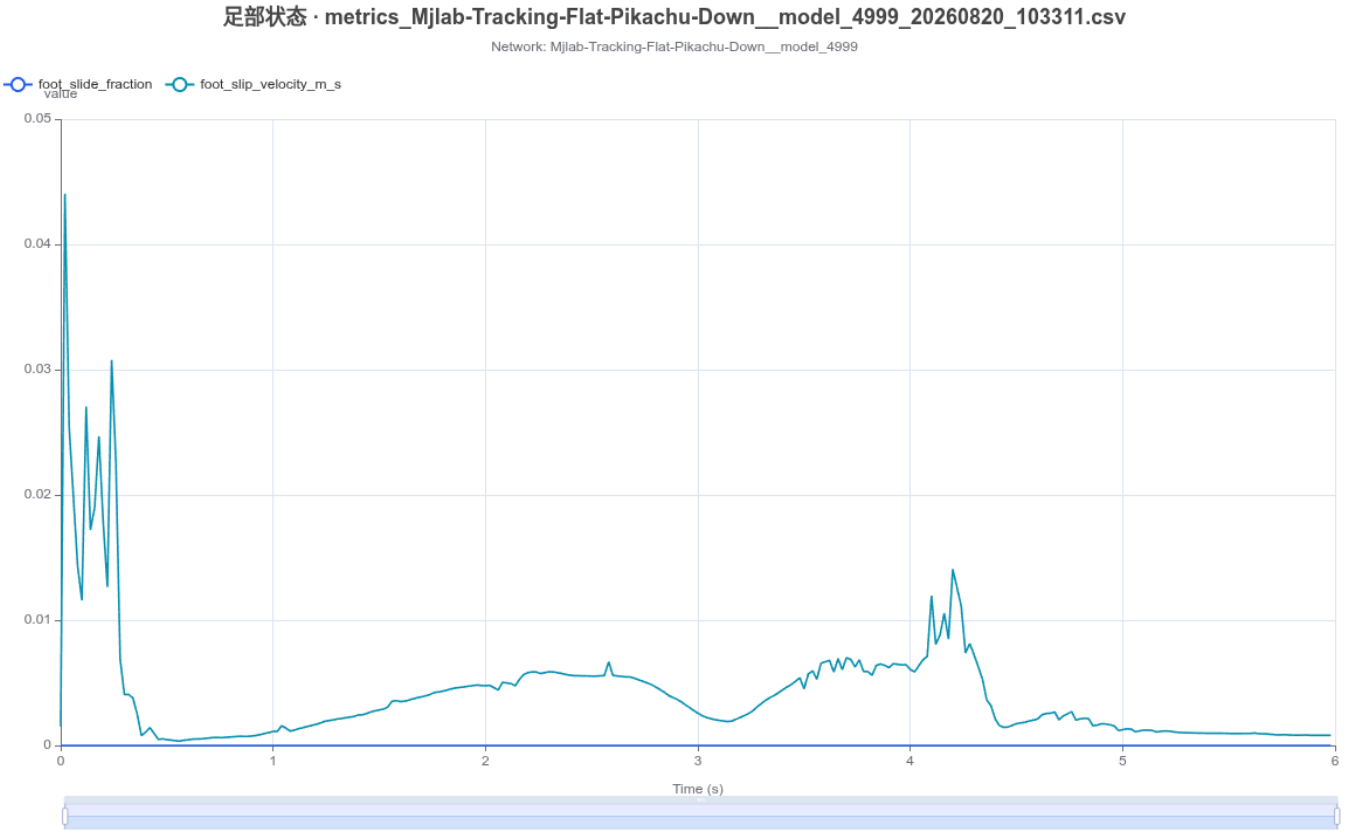
机体状态



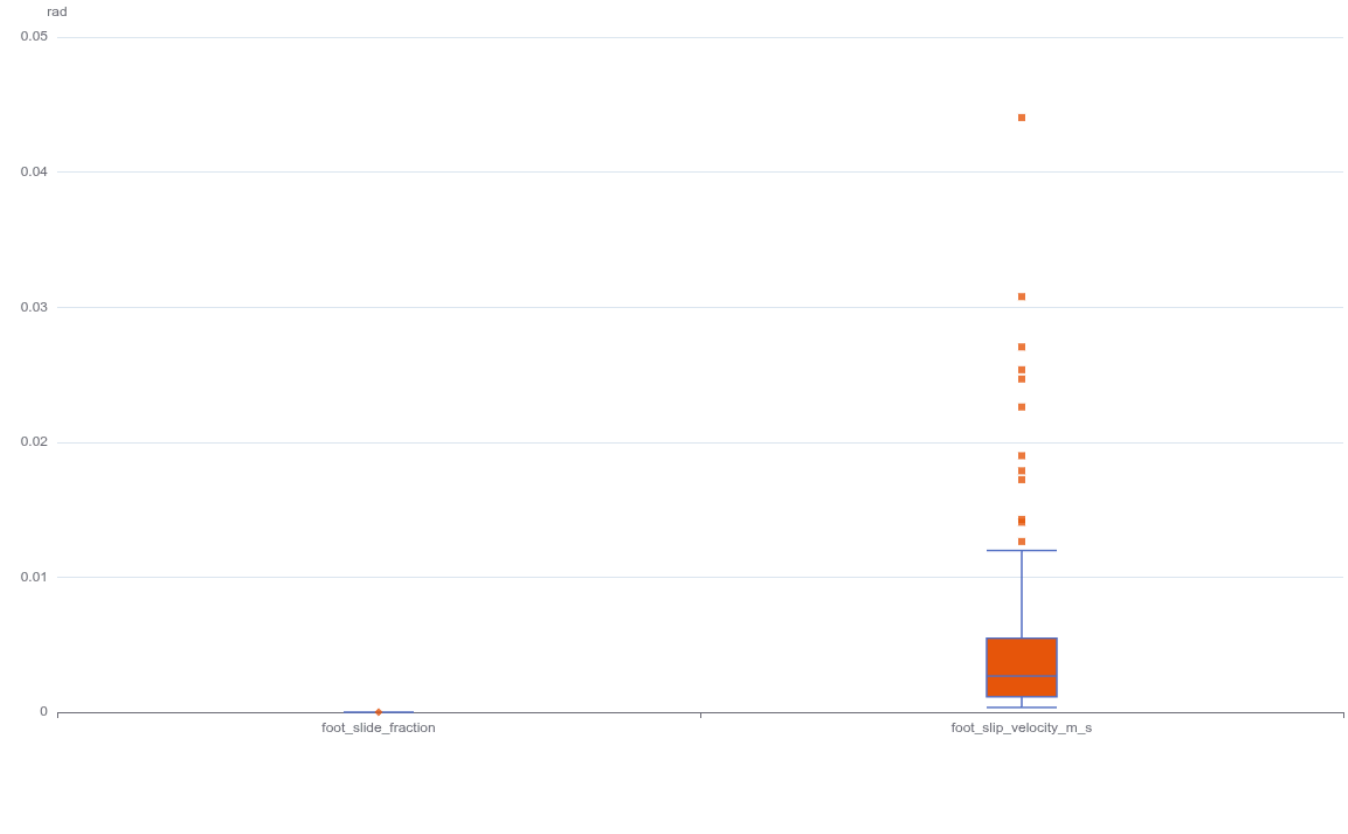
Metric	Vel限	Torq限	N	Min	Max	Mean	RMS	Peak |x|	P95 |x|
base_height_m	–	–	277	0.160632	0.180606	0.168230	0.168320	0.180606	0.177078



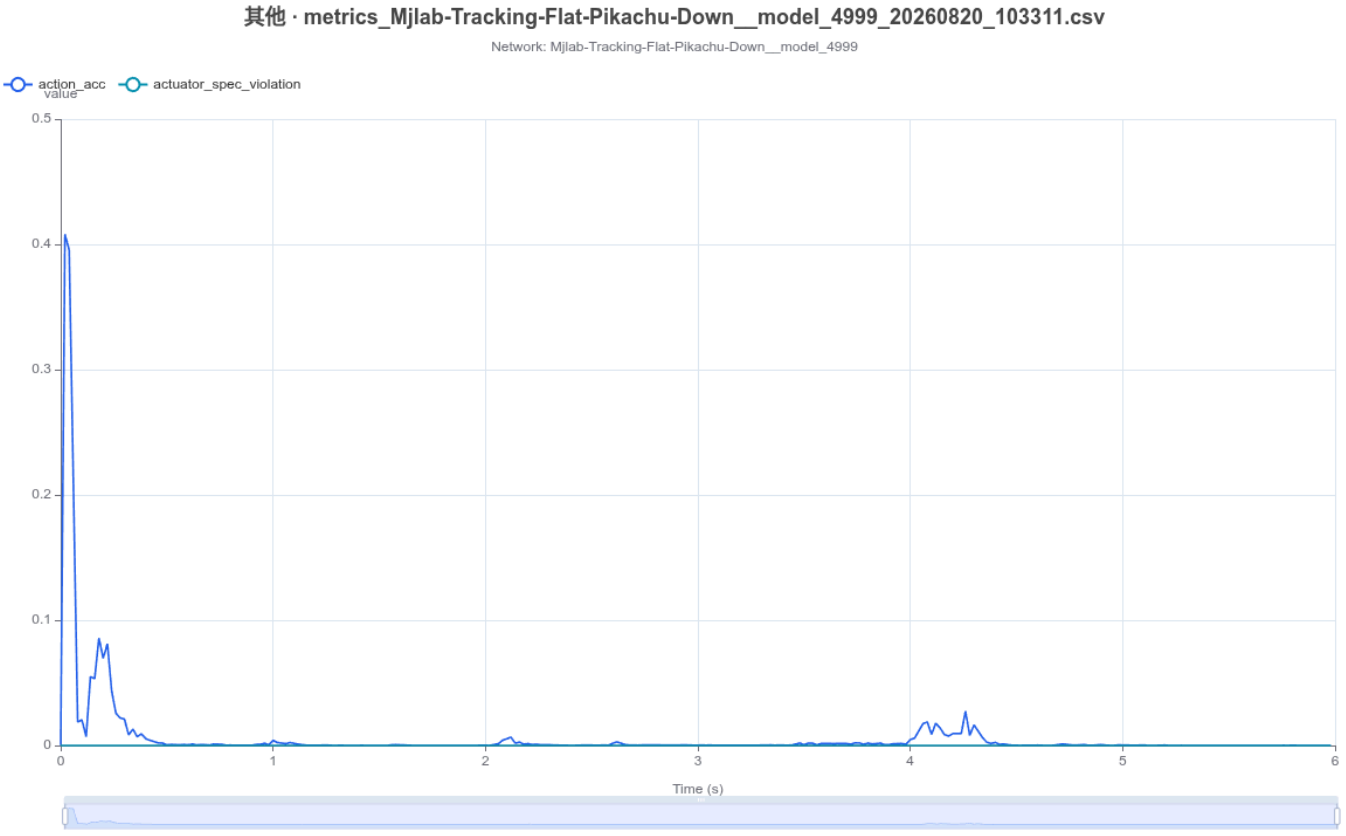
足部状态



Metric	Vel限	Torq限	N	Min	Max	Mean	RMS	Peak |x|	P95 |x|
foot_slide_fraction	–	–	277	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000
foot_slip_velocity_m_s	–	–	277	0.000352634	0.0440393	0.00405917	0.00633616	0.0440393	0.0111163



其他



Metric	Vel限	Torq限	N	Min	Max	Mean	RMS	Peak |x|	P95 |x|
action_acc	–	–	277	0.0000122307	0.408132	0.00602558	0.0358125	0.408132	0.0188611
actuator_spec_violation	–	–	277	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000

